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RAPID ACTIVE CLEARANCE CONTROL SYSTEM OF INTER STAGE AND MID-SEALS

Abstract

Example apparatus, systems, and methods for rapid active clearance control of inter-stage and mid-stage seals are disclosed. An example apparatus to control clearance for a turbine engine comprises a case surrounding at least part of the turbine engine and defining an opening therethrough; a nozzle, the nozzle including a reference pressure sensor and a static pressure sensor on a tip of the nozzle; an actuator including a multilayer stack of material, a rod coupled to the first actuator and coupled to the nozzle through the opening in the case, the rod to move the nozzle based on contraction or expansion of the multilayer stack of material; and a controller to calculate and set the clearance between the rotor and the nozzle by supplying an electrical current to the multilayer stack to cause the multilayer stack to at least one of expand or contract.

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Background/Summary

RELATED APPLICATIONS [0001] This patent arises from a continuation of U.S. patent application Ser. No. 18/337,321 (now U.S. Patent _____), filed on Jun. 19, 2023. U.S. patent application Ser. No. 18/337,321 is hereby incorporated herein by reference in its entirety. Priority to U.S. patent application Ser. No. 18/337,321 is hereby claimed.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to a gas turbine engine, and, more particularly, to rapid active clearance control systems of inter stage and mid-seals of a gas turbine engine.

BACKGROUND

[0003] A gas turbine engine generally includes, in serial flow order, an inlet section, a compressor section, a combustion section, a turbine section, and an exhaust section. In operation, air enters the inlet section and flows to the compressor section where one or more axial compressors progressively compress the air until it reaches the combustion section. Fuel mixes with the compressed air and burns within the combustion section, thereby creating combustion gases. The combustion gases flow from the combustion section through a hot gas path defined within the turbine section and then exit the turbine section via the exhaust section.

[0004] In general, it is desirable for a gas turbine engine to maintain clearance between the tip of a blade in the gas turbine engine and the stationary parts of the gas turbine engine (e.g., the gas turbine engine casing, stator, etc.). During operation, the gas turbine engine is exposed to thermal (e.g., hot and cold air pumped into the gas turbine engine, etc.) and mechanical loads (e.g., centrifugal force on the blades on the gas turbine engine, etc.), which can expand and contract the gas turbine engine casing and rotor. The expansion and contraction of the gas turbine engine casing can control the clearance between the blade tip and the stationary parts of the gas turbine engine. There is a continuing need to control the clearance between the blade tip and the engine casing that fluctuates during normal operation for a gas turbine engine to avoid damage to the gas turbine engine (e.g., wear, breakage, etc. due to unintentional rub) and control the clearances for better engine performance and operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a schematic cross-sectional view of an example gas turbine engine in accordance with the examples disclosed herein.

[0006] FIG. 2 is a schematic cross-sectional view of an example gas turbine engine with a conventional active clearance control (ACC) system.

[0007] FIG. 3 is a schematic cross-sectional view of an ACC system for a high pressure turbine engine.

[0008] FIG. 4 is a schematic cross-sectional view of an ACC system for a low pressure turbine engine.

[0009] FIG. 5 is a schematic cross-sectional view of a second example ACC system for a high pressure turbine engine in accordance with teachings disclosed herein.

[0010] FIG. 6 is a schematic cross-sectional view of a third example ACC system for a high pressure turbine engine in accordance with teachings disclosed herein.

[0011] FIG. 7A is an example conversion curve determined using clearance and pressure efficiency for a new blade.

[0012] FIG. 7B is an example conversion curve determined using clearance and pressure efficiency for a blade with tip loss.

[0013] FIG. 8A illustrates an example conversion curve determined using clearance and pressure efficiency for both a new blade and a blade with tip loss during high power operation.

[0014] FIG. 8B illustrates an example conversion curve determined using clearance and pressure efficiency for both a new blade and a blade with tip loss during low power operation.

[0015] FIG. 9A illustrates an example one-point pressure measurement at a mid-point.

[0016] FIG. 9B illustrates an example one-point pressure measurement at an aft-point.

[0017] FIG. 10A illustrates a two-point pressure measurement method at forward and mid points.

[0018] FIG. 10B illustrates a two-point pressure measurement method at forward and aft points.

[0019] FIG. 11A illustrates a first three-point pressure measurement method at forward, mid, and aft points.

[0020] FIG. 11B illustrates a second three-point pressure measurement method at forward, mid, and aft points.

[0021] FIG. 12 is a schematic cross-sectional view of a second example ACC system for a low pressure turbine engine in accordance with teachings disclosed herein.

[0022] FIG. 13 is a schematic cross-sectional view of a third example ACC system for a low pressure turbine engine in accordance with teachings disclosed herein.

[0023] FIG. 14 is a block diagram of an Active Clearance Controller which can implement the examples disclosed herein.

[0024] FIG. 15 is a block diagram of the clearance determination circuitry of FIG. 14.

[0025] FIG. 16 illustrates a flowchart representative of example machine readable instructions which may be executed to implement the example clearance determination circuitry of FIG. 15.

[0026] FIG. 17 is a block diagram of an example processing platform structured to execute the instructions of FIG. 16 to implement the example clearance determination instructions of FIG. 15.

[0027] FIG. 18 is a schematic cross-sectional view of a fourth example ACC system for a high pressure turbine engine in accordance with teachings disclosed herein.

[0028] The figures are not to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts.

[0029] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another. As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0030] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation

to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0031] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name. As used herein, “approximately” and “about” refer to dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections.

DETAILED DESCRIPTION

[0032] In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific examples that may be practiced. These examples are described in sufficient detail to enable one skilled in the art to practice the subject matter, and it is to be understood that other examples may be utilized. The following detailed description is, therefore, provided to describe example implementations and not to be taken limiting on the scope of the subject matter described in this disclosure. Certain features from different aspects of the following description may be combined to form yet new aspects of the subject matter discussed below.

[0033] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc. may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

[0034] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” entity, as used herein, refers to one or more of that entity. The terms “a” (or “an”), “one or more”, and “at least one” can be used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., a single unit or processor. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or

advantageous.

[0035] As used herein, the terms “system,” “unit,” “module,” “engine,” etc., may include a hardware and/or software system that operates to perform one or more functions. For example, a module, unit, or system may include a computer processor, controller, and/or other logic-based device that performs operations based on instructions stored on a tangible and non-transitory computer readable storage medium, such as a computer memory. Alternatively, a module, unit, engine, or system may include a hard-wired device that performs operations based on hard-wired logic of the device. Various modules, units, engines, and/or systems shown in the attached figures may represent the hardware that operates based on software or hardwired instructions, the software that directs hardware to perform the operations, or a combination thereof.

[0036] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. As used herein, “vertical” refers to the direction perpendicular to the ground. As used herein, “horizontal” refers to the direction parallel to the centerline of the gas turbine engine **100**. As used herein, “lateral” refers to the direction perpendicular to the axial and vertical directions (e.g., into and out of the plane of FIGS. **1**, **2**, etc.).

[0037] In some examples used herein, the term “substantially” is used to describe a relationship between two parts that is within three degrees of the stated relationship (e.g., a substantially collinear relationship is within three degrees of being linear, a substantially perpendicular relationship is within three degrees of being perpendicular, a substantially parallel relationship is within three degrees of being parallel, etc.).

[0038] A turbine engine, also called a combustion turbine or a gas turbine, is a type of internal combustion engine. Turbine engines are commonly utilized in aircraft and power-generation applications. As used herein, the terms “asset,” “aircraft turbine engine,” “gas turbine,” “land-based turbine engine,” and “turbine engine” are used interchangeably. A basic operation of the turbine engine includes an intake of fresh atmospheric air flow through the front of the turbine engine with a fan. In some examples, the air flow travels through an intermediate-pressure compressor, or a booster compressor located between the fan and a high-pressure compressor. A turbine engine also includes a turbine with an intricate array of alternating rotating and stationary airfoil-section blades. As the hot combustion gas passes through the turbine, the hot combustion gas expands, causing the rotating blades to spin.

[0039] The components of the turbine engine (e.g., the fan, the booster compressor, the high-pressure compressor, the high-pressure turbine, the low-pressure turbine, etc.) can degrade over time due to demanding operating conditions such as extreme temperature and vibration. During operation, the turbine engine components are exposed to thermal (e.g., hot and cold air pumped into the turbine engine, etc.) and mechanical loads (e.g., centrifugal force on the blades on the turbine engine, etc.), which can expand and contract the turbine engine casing and/or compressor casing within the turbine engine along with other components of the turbine engine and/or its compressor. The expansion and contraction of the turbine engine casing and/or compressor casing within the turbine engine can change the clearance between the blades' tips and the stationary components of the turbine engine. In some examples, if the clearance between the blades' tips and the stationary components is not controlled, then the blades' tips and stationary components can collide during operation and lead to further degradation of the components of the turbine engine.

[0040] An Active Clearance Control (ACC) System was developed to improve engine performance by managing the clearance between a gas turbine containment structure and a tip of a rotating blade without unexpected harmful rub events during flight and ground operations. A conventional ACC System includes using cooling air from a fan or compressor to control the clearance between the blade tip and an engine component that has shrunk (e.g., the stator, the case, etc.). The conventional ACC system is limited in that clearance is only modulated in one direction (e.g., engine component

shrinkage). For a hot rotor condition (e.g., the engine component(s) are expanded), the conventional ACC system waits for rotor-stator thermal/mechanical growth matching to escape the hot rotor condition (e.g., modulate the blade tip clearance). Tip clearance is maintained at a minimum value to ensure maximum propulsive efficiency. For example, combusted gas temperatures can exceed 1,000 degrees Celsius, causing turbine blade expansion as well as expansion of the containment structure, increasing tip clearance and reducing overall turbine efficiency (e.g., increased fuel burn and fuel consumption). Control of thermal expansion and contraction of the containment structure permits turbine tip clearance control.

[0041] Conventional ACC systems have an inability to directly control a stator nozzle (also referred to as a vane) and a connected compressor inter-stage seal. The compressor inter-stage seal is passively dependent on the ACC. The ACC is not connected to the compressor, meaning that the nozzle conventionally hangs connected between two hangers, with a mid-stage seal, that are connected to the case. The hangers have shrouds attached that are affected by temperature and pressures from the blades, which may sometimes cause uneven displacement between the forward and aft sides. This uneven displacement results in nozzle rocking, causing pressure loss at both the inter-stage seal and the mid-stage seal.

[0042] The inter-stage seal clearance can create problems regarding pressure balancing on forward and aft sides. If the clearance is too open, temperature has a tendency to build up on the forward side by the blade, resulting in airflow and pressure loss to the aft side. The change in pressures alters the displacement of the hanger and shroud, resulting in nozzle rocking and further pressure loss.

[0043] In the instance of temperature build-up, the heat can cause thermal expansion radially inwardly of the case and connected components. The inter-stage seal clearance closes as a result of the expansion. With current, conventional ACC systems, there is a time delay to open the inter-stage seal clearance because the ACC only has the ability to control motion inward, towards the blades.

[0044] Examples disclosed herein improve an ACC system using actuator(s) with a multilayer stack of piezoelectric material (also referred to herein as a multilayer stack, piezoelectric material or piezoelectric stack) that provide rapid active clearance control of the inter stage and mid-seals without the mechanical delay seen in the conventional ACC system. Examples disclosed herein maintain desired clearances between the inter-stage seal and rotor without additional margin for various operating conditions, which leads to performance improvement and provide better exhaust gas temperature (EGT) control capability. In certain examples, the multilayer stack generates linear displacement when an electric current is applied. The linear displacement can have a force, and examples disclosed herein apply the linear force of the multilayer stack (made of piezoelectric material) for the ACC system to achieve rapid active clearance control of the inter stage and mid-seals. Examples disclosed herein apply the mechanical force from the linear displacement of the multilayer stack on to modulating the ACC system. Examples disclosed herein can include other materials that generate linear displacement such as, shape memory alloy (SMA), etc. The range of displacement is increased by adding layers of piezoelectric material or SMA, where more layers in a stack provides more radial movement range and gives the ACC system more muscle capability.

[0045] Examples disclosed herein use an actuator to house the multilayer stack. The actuator achieves clearance in two directions (e.g., radially inward and outward). Examples disclosed herein do not need additional clearance margin for maximum transient closure or hot-rotor condition like the conventional ACC system. Examples disclosed herein provide significant specific fuel consumption (SFC) improvement on tighter clearance and a better EGT control as there are no additional margins for transient closure or the hot rotor condition.

[0046] An example actuator design is a direct linear square actuator that is a tube in a piston style. The example actuator can be amplified if more muscle is necessary. The range and requirements depend from module to module, however, the force associated with the example actuator is in the

range of about 450 to about 700 pounds-force. The example stroke/muscle is in the range of about 5 to about 14 mils. The example operating temperature range is about 120 to about 250 degrees Fahrenheit. The example actuator modulates a 1-mil derivative with a response time of approximately one millisecond. In an alternate design, the actuator is circular/disc shaped.

[0047] In the examples disclosed herein, using the actuator in conjunction with the multilayer stack can provide the flexibility to implement many different casing designs with compact and simple piezo stacks while providing the same high mechanical force as a conventional ACC.

[0048] Certain examples provide an engine controller, referred to as a full authority digital engine (or electronics) control (FADEC). The FADEC includes a digital computer, referred to as an electronic engine controller (EEC) or engine control unit (ECU), and related accessories that control aspects of aircraft engine performance. The FADEC can be used with a variety of engines such as piston engines, jet engines, other aircraft engines, etc. In certain examples, the EEC/ECU is provided separate from the FADEC, allowing manual override or intervention by a pilot and/or other operator.

[0049] In examples disclosed herein, the engine controller receives values for a plurality of input variables relating to flight condition (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.). The engine controller computes engine operating parameters such as fuel flow, stator nozzle position, air bleed valve position, etc., using the flight condition data. The engine operating parameters can be used by the engine controller to control operation of the multilayer stack to modulate blade tip and seal clearances in the turbine engine.

[0050] Reference now will be made in detail to embodiments of the presently disclosed technology, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the presently disclosed technology, not limitation of the presently disclosed technology. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the presently disclosed technology without departing from the scope or spirit of the presently disclosed technology. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the presently disclosed technology covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0051] FIG. 1 is a schematic cross-sectional view of a conventional turbofan-type gas turbine engine **100**. As shown in FIG. 1, the gas turbine engine **100** defines a longitudinal or axial centerline axis **102** extending therethrough for reference. In general, the gas turbine engine **100** may include a core turbine **104** disposed downstream from a fan section **106**.

[0052] The core turbine **104** generally includes a substantially tubular outer casing **108** that defines an annular inlet **110**. The outer casing **108** can be formed from a single casing or multiple casings. The outer casing **108** encloses, in serial flow relationship, a compressor section having a booster or low pressure compressor (“LP compressor **112**”) and a high pressure compressor (“HP compressor **114**”), a combustion section **116**, a turbine section having a high pressure turbine (“HP turbine **118**”) and a low pressure turbine (“LP turbine **120**”), and an exhaust section **122**. A high pressure shaft or spool (“HP shaft **124**”) drivingly couples the HP turbine **118** and the HP compressor **114**. A low pressure shaft or spool (“LP shaft **126**”) drivingly couples the LP turbine **120** and the LP compressor **112**. The LP shaft **126** may also couple to a fan spool or shaft (“fan shaft **128**”) of the fan section **106**. In some examples, the LP shaft **126** may couple directly to the fan shaft **128** (e.g., a direct-drive configuration). In alternative configurations, the LP shaft **126** may couple to the fan shaft **128** via a reduction gearbox **130** (e.g., an indirect-drive or geared-drive configuration).

[0053] As shown in FIG. 1, the fan section **106** includes a plurality of fan blades **132** coupled to and extending radially outwardly from the fan shaft **128**. An annular fan casing or nacelle **134** circumferentially encloses the fan section **106** and/or at least a portion of the core turbine **104**. The nacelle **134** is supported relative to the core turbine **104** by a plurality of circumferentially spaced apart outlet guide vanes **136**. Furthermore, a downstream section **138** of the nacelle **134** can

enclose an outer portion of the core turbine **104** to define a bypass airflow passage **140** therebetween.

[0054] As illustrated in FIG. **1**, air **142** enters an inlet portion **144** of the gas turbine engine **100** during operation thereof. A first portion **146** of the air **142** flows into the bypass airflow passage **140**, while a second portion **148** of the air **142** flows into the inlet **110** of the LP compressor **112**. One or more sequential stages of LP compressor stator nozzles **150** and LP compressor rotor blades **152** coupled to the LP shaft **126** progressively compress the second portion **148** of the air **142** flowing through the LP compressor **112** en route to the HP compressor **114**. Next, one or more sequential stages of HP compressor stator nozzles **154** and HP compressor rotor blades **156** coupled to the HP shaft **124** further compress the second portion **148** of the air **142** flowing through the HP compressor **114**. This provides compressed air **158** to the combustion section **116** where it mixes with fuel and burns to provide combustion gases **160**.

[0055] The combustion gases **160** flow through the HP turbine **118** in which one or more sequential stages of HP turbine stator nozzles **162** and HP turbine rotor blades **164** coupled to the HP shaft **124** extract a first portion of kinetic and/or thermal energy from the combustion gases **160**. This energy extraction supports operation of the HP compressor **114**. The combustion gases **160** then flow through the LP turbine **120** where one or more sequential stages of LP turbine stator nozzles **166** and LP turbine rotor blades **168** coupled to the LP shaft **126** extract a second portion of thermal and/or kinetic energy therefrom. This energy extraction causes the LP shaft **126** to rotate, thereby supporting operation of the LP compressor **112** and/or rotation of the fan shaft **128**. The combustion gases **160** then exit the core turbine **104** through the exhaust section **122** thereof.

[0056] Along with the gas turbine engine **100**, the core turbine **104** serves a similar purpose and sees a similar environment in land-based gas turbines, turbojet engines in which the ratio of the first portion **146** of the air **142** to the second portion **148** of the air **142** is less than that of a turbofan, and unducted fan engines in which the fan section **106** is devoid of the nacelle **134**. In each of the turbofan, turbojet, and unducted engines, a speed reduction device (e.g., the reduction gearbox **130**) may be included between any shafts and spools. For example, the reduction gearbox **130** may be disposed between the LP shaft **126** and the fan shaft **128** of the fan section **106**.

[0057] FIG. **2** is a schematic cross-sectional view of an example gas turbine engine with a conventional active clearance control (ACC) system **200**. The ACC system **200** includes an example main pipe **205**, an example high pressure turbine **210**, an example low pressure turbine **215**, example manifolds **220A**, **220B**, **220C**, example flanges **225A**, **225B**, and example mid-rings **230A**, **230B**. In the illustrated example of FIG. **2**, air from a fan (e.g., from the fan section **106**) enters the main pipe **205**, where the airflow in the main pipe **205** is shown by the arrows in FIG. **2**. In some examples, the inlet of the main pipe **205** is located at the fan (e.g., the fan section **106** of FIG. **1**) or upstream of a compressor (e.g., the HP compressor **114** of FIG. **1**) for the high pressure turbine **210**. In some examples, the ACC system **200** is applicable for a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. **1**) and the low pressure turbine **215**. The main pipe **205** delivers the air from the fan to the manifolds **220A**, **220B**, **220C**. The manifolds **220A**, **220B**, **220C** evenly distribute the air from the fan to the high pressure turbine **210** and the low pressure turbine **215**. In some examples, the high pressure turbine **210** is substantially similar to the HP turbine **118**, and the low pressure turbine **215** is substantially similar to the LP turbine **120**. The flanges **225A**, **225B** and mid-rings **230A**, **230B** are joined to the outer surfaces of the high pressure turbine **210** case and the low pressure turbine **215** case. The flanges **225A**, **225B** and mid-rings **230A**, **230B** are configured to contract radially inward and/or expand radially outward in responses to changes in temperature (e.g., changes in temperature caused by the air from the manifolds **220A**, **220B**, **220C**). In some examples, at least some of the air is directed to impinge on the surfaces of the flanges **225A**, **225B** and mid-rings **230A**, **230B**. In some examples, the contraction inward and expansion outward of the flanges **225A**, **225B** and the mid-rings **230A**, **230B** can change blade tip clearances in the high pressure turbine **210** and the low pressure turbine **215**.

[0058] In the following examples, EGT refers to a temperature of turbine exhaust gases during exit from the turbine unit, the temperature measured using thermocouples mounted in the exhaust stream. Active clearance control maintains optimal or otherwise improved clearance in part to help ensure that EGT remains below its limit (e.g., a temperature threshold), which improves engine efficiency and time-on-wing. Likewise, tighter blade tip clearances are maintained to reduce air leakage over blade **164**, **168** tips, otherwise rotor inlet temperatures are increased to achieve the same level of performance and hot section components experience a reduced life cycle due to the temperature increases (e.g., thermal fatigue) to produce the same amount of work. Furthermore, maintenance costs can be reduced by ensuring engine efficiency through optimized tip clearances via ACC.

[0059] FIG. **3** is a schematic cross-sectional view of a prior ACC system **300** for an example high pressure turbine, such as the gas turbine engine **100** of FIG. **1**. The prior ACC system **300** includes a case **305**, guiding hooks **310A**, **310B**, a hanger **315**, a shroud **320**, a blade **325**, mid stage seals **307**, a stator nozzle **335**, and an inter-stage seal **340**. In the illustrated example of FIG. **3**, the case **305** is the casing surrounding either the HP turbine **118**, the LP turbine **120**, and/or the compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. **1**). The case **305** includes the guiding hooks **310A**, **310B**, and the guiding hooks **310A**, **310B** connect the case **305** to the hanger **315**. The hanger **315** is connected to the shroud **320**. The stator nozzle **335** is also connected to the hangers **315** with an inter-stage seal **340**. Mid stage seals **307** balance pressures of the stator nozzles **335** of a forward and an aft cavity.

[0060] In the illustrated example of FIG. **3**, the prior ACC system **300** determines the clearance between the shroud **320** and the blade **325**, as well as the inter-stage seal **340** and the rotor **345**. The prior ACC system **300** uses the ACC **330** to control the movement of the shroud **320** and stator nozzle **335** in only one direction (e.g., inward towards the blade **325**). The prior ACC system **300** uses cooling airflow from the compressor or fan (shown in FIG. **2**) to cool the case **305**. The case **305** shrinks (e.g., moves inward) as it is cooled by the airflow. The case **305** moves the hanger **315**, shroud **320**, and stator nozzle **335** inward towards the blade **325**. The prior ACC system **300** is unable to move the case **305**, the hanger **315**, the shroud **320**, and the stator nozzle **335** for expansion. For example, the ACC system **300** is unable to expand the case **305** (e.g., move outward) to increase the clearance between the shroud **320** and the blade **325** or between the inter-stage seal **340** and the rotor **345**. In such examples, the prior ACC system **300** waits for clearance between the shroud **320** and the blade **325** or between the inter-stage seal **340** and the rotor **345** to open (e.g., increase). The prior ACC system **300** does not provide bi-directional control of the clearance between the shroud **320** and the blade **325** or subsequently between the inter-stage seal **340** and the rotor **345**.

[0061] In some examples (e.g., the prior ACC system **300** of FIG. **3**), an ACC system directs airflow around the case of an engine to control clearance between the case and the blade tip, as well as the inter-stage seal and the rotor. For example, the ACC system controls the cooling airflow (shown in FIG. **2**) from a compressor or fan to the case **305**. In some examples, the ACC system mixes hot and cold air from a compressor and a bypass duct (contains gas turbine engine airflow that bypassed the engine core) respectively to a desired temperature. In some examples, the ACC system helps to maintain and adjust the clearance between the inter-stage seal and the rotor in prior ACC systems. In prior ACC systems (e.g., the prior ACC system **300** of FIG. **3**), cooling airflow around the engine case (e.g., case **305**) adjusts the clearance by controlling the thermal expansion and contraction of the case. In some examples, the ACC **330** controls the cooling airflow to contract the turbine engine case. For example, the prior ACC system **300** directs cooling airflow to the case **305** to contract the case **305** and restricts the cooling airflow to the case **305** to expand the case **305**. The ACC system **300** controls the cooling airflow to adjust the clearance to compensate any changes in the blade of the turbine engine. In some examples, the ACC system **300** is controlled by a controller in the turbine engine (e.g., the FADEC). The FADEC sends electrical

control signals to the ACC system **300** to signal the ACC system **300** to modulate the airflow to control the case thermal expansion. The ACC system **300** ultimately controls the amount of cooling airflow to manage the turbine engine casing temperatures, thereby adjusting the inter-stage seal clearance. The clearance control fails to be performed in real-time and is without finite control over the contraction or expansion of the case **305** or any components with displacements dependent therefrom.

[0062] FIG. **4** is a schematic cross-sectional view of an example ACC system **400** in a low pressure turbine (LPT) implementation. With this prior LPT ACC system **400**, a case **410** is connected directly to a shroud **420** and a stator nozzle **430**. Airfoils **425** are on either side of the stator nozzle **430**. Heat shields **415** are in place to mitigate thermal expansion of the case when pressure builds up. Similar to the ACC system in FIG. **3**, the ACC **405** directs airflow around the case **410** of an engine to control clearance between the case **410** and the blade tip, as well as the inter-stage seal and the rotor. For example, the ACC **405** controls the cooling airflow (shown in FIG. **2**) from a compressor or fan to the case **410**. The clearance control fails to be performed in real-time and is without finite control over the contraction or expansion of the case **410** or any components with displacements dependent therefrom.

[0063] FIG. **5** is a schematic cross-sectional view of an example high pressure turbine ACC system **500** in accordance with teachings disclosed herein. The example ACC system **500** of FIG. **5** includes an actuator **550**, a rod **560**, a mid-stage seal **517**, a case **505**, guide hooks **510A**, **510B**, a hanger **515**, a shroud **520**, a blade **525**, a stator nozzle **535**, a placement seal **507** in the case **505** to allow for placement of the rod **560**, and an inter-stage seal **540**. The actuator **550** includes a multilayer stack **555**, for example. In this example, forward pressure sensor **565** and aft pressure sensor **570** are attached to the inter-stage seal **540** to measure pressure and send the measurements to the active clearance controller **575**. The active clearance controller **575** integrates the feedback from the forward and aft sensors **565**, **570** with the actuator movement to enable finite control of the displacement of the stator nozzle **535** and inter-stage seal **540**. The example ACC system **500** of FIG. **5** includes an open clearance between the inter-stage seal **540** and the rotor **545**.

[0064] The active clearance controller **575** integrates the feedback from the forward and aft sensors **565**, **570** with the actuator movement and can be set to be either closed or open loop. In both instances, the active clearance controller **575** accounts for not only the positioning of the stator nozzle **535** and inter-stage seal **540** with respect to the rotor **545**, but also the hanger **515** and shroud **520** with respect to the blade **525** on either side of the stator nozzle **535**. The active clearance controller accounts for varying pressures and temperatures that create a flow from upstream (relatively higher pressure) to downstream (relatively lower pressure) on either side of the stator nozzle. A balance is maintained between the clearance allowed by inter-stage seal **540** and the rotor **545**, the aft-side pressure between aft shroud **520b** and aft blade **525b**, and the forward-side pressure between forward shroud **520a** and forward blade **525a**. The pressure, and subsequently the clearance, is measured by an aft pressure sensor **570** and a forward pressure sensor **565**. The active clearance controller **575** accounts for the pressure, the clearance, blade tip loss, nozzle rocking, and other engine parameters such as the power application, the altitude, etc. to adjust all actuators **550** and multilayer stacks **555**. As each actuator **550** and multilayer stack **555** is connected to the rod **560** and subsequently the stator nozzle **535**, the active clearance controller **575** has finite control radially inward and outward over the clearance between the inter-stage seal **540** and the rotor **545**. Additionally, the use of piezoelectric material for multilayer stack **555** enables substantially real-time rapid response.

[0065] In a closed loop control system, a clearance calculation is utilized, where a target clearance is set. The actual clearance is calculated and compared to the target clearance. The clearance calculation includes an engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure to calculate the actual clearance. The actuator **550** with the multilayer stack **555** is then manipulated to achieve the target clearance. The calculation and

actuator manipulation are performed in substantially real-time.

[0066] In an instance where the active clearance controller uses an open loop system to control the clearance, conversion curves are used to correlate a normalized pressure measurement with a clearance measurement (example conversion curves are provided in connection with FIGS. 7A-8B). First, the pressure is obtained from the pressure sensors 565, 570. Once the pressure is measured, the pressure is normalized and the conversion curve is used to calculate the clearance. The clearance is compared to a predetermined set value for clearance. The actuator 550 and multilayer stack 555 are then signaled by the active clearance controller to achieve the target by moving the associated rod 560. The measurement, comparison and actuation are performed in substantially real-time to help ensure that proper clearance is maintained and rub/fragmentation events are avoided.

[0067] FIG. 6 shows an alternative implementation of an ACC system 600. The example ACC system 600 of FIG. 6 includes an active clearance controller 675, an actuator 650, a rod 660, a case 605, a hanger 615, a shroud 620, a blade 625, a stator nozzle 635, a mid-stage seal 617, a placement seal 607 to enable placement of the rod 660 through the case 605, and an inter-stage seal 640. The actuator 650 of FIG. 6 includes the multilayer stack 655, which is expanded (or elongated) in the radial direction and contracted in the axial direction. In examples disclosed herein, the case 605 includes the guiding hooks 610A, 610B, and the guiding hooks 610A, 610B connect the case 605 to the hanger 615. The hanger 615 is connected to the shroud 620. Also attached to the hangers is the stator nozzle 635 with the inter-stage seal 640 at the tip. Pressure sensors 665, 670 are on the forward and aft sides of the inter-stage seal 640. In this example, two actuator 650 and multilayer stack 655 sets are used to control the radial actuation of the stator nozzle 635 and the inter-stage seal 640 to control the clearance between the inter-stage seal 640 and rotor 645.

[0068] Stator nozzle rocking causes pressure imbalance. The pressure imbalance decreases the effectiveness of the seal, subsequently causing flow and deeper rub than design intent, which changes the thermal conditions around seals and affects part life. There is an increased associated risk of potential part failure as the rocking and rub continues. Additionally, nozzle rocking causes flow path step unbalancing, which impacts aero efficiency. The utilization of two actuators and multilayer stacks connected to the stator nozzle 635 provides further control over nozzle rocking, which is uneven displacement of the forward and aft sides of the stator nozzle 635 and inter-stage seal 640. Equal displacement of the forward and aft sides of the inter-stage seal 640 yields improved control over airflow from the forward side to the aft side, resulting in less difficulty controlling a temperature and pressure differential, as relative to a stator nozzle 635 subject to nozzle rocking. The two actuators 650 and piezoelectric stacks 655 with rods 660 connected to the stator nozzle 635 are used in conjunction with actuators 650, multilayer stacks 655 and rods 660 connected to the hangers 615 and shrouds 620 to give complete control of temperature, pressure, and blade tip loss. The result is improvement in maintenance over the aero efficiency, maintenance of thermal conditions around the inter-stage seal 640, as well as prevention of negatively impacted part life.

[0069] In the illustrated examples of FIGS. 5 and 6, the actuator 550, 650 is located outside of the case 505, 605, so that the case 505, 605 encloses all components except the actuator 550, 650 and the active clearance controller 575, 675. In some examples, the case 505, 605 is a case surrounding a high pressure turbine (e.g., the HP turbine 118 of FIG. 1), a low pressure turbine (e.g., the LP turbine 120 of FIG. 1), and/or a compressor (e.g., the HP compressor 114 and LP compressor 112 of FIG. 1). In some examples, locating the actuator 550, 650 outside of the case 505, 605 prevents material temperature limitations from affecting the actuator 550, 650 by locating the insulation of the actuator box to preserve thermal condition. For example, hot gas temperatures in a high pressure turbine such as the HP turbine 118 of FIG. 1, can cause material limitations for the actuator 550, 650 if the actuator 550, 650 was located inside the case 505, 605. In the example

ACC systems **500** and **600**, the actuator **550**, **650** includes a multilayer stack of piezoelectric material **555**, **655**. In some examples, the piezoelectric material of the multilayer stack **555**, **655** includes quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement such as, shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included. In some examples, locating the actuator **550**, **650** and the multilayer stack **555**, **655** outside of the case **505**, **605** helps to preserve the piezoelectric material in a cold condition without concern of temperature limitations. The location of the actuator **550**, **650** and the multilayer stack **555**, **655** provides a benefit of easy access for maintenance and part replacement, for example.

[0070] Shape memory alloy materials are additionally and/or alternatively used to generate linear displacement. The insulation of actuator **550**, **650** extends the limit of the thermal condition of the shape memory alloy materials in the actuators. The shape memory alloy materials are deformed based on the thermal condition. The thermal condition is controlled based on the electrical power supplied to the actuator **550**, **650**.

[0071] In the illustrated examples of FIGS. **5** and **6**, the multilayer stack **555**, **655** is connected to the rod **560**, **660**. The rod **560**, **660** is connected to the stator nozzle **635** and inter-stage seal **640**. Since the actuator **550**, **650** and the multilayer stack **555**, **655** are located outside of the case **505**, **605**, the rod **560**, **660** is inserted through the case **505**, **605** to connect to the multilayer stack **555**, **655** and the stator nozzle **535**, **635**. In some examples, the opening in the case **505**, **605** for the rod **560**, **660** to be inserted through introduces possible leakage through the case **505**, **605**. In such examples, the rod **560**, **660** is surrounded by the placement seal **507**, **607** to seal the opening in the case **505**, **605** that the rod **560**, **660** is inserted through. In one example, the placement seal **507**, **607** is a W-seal. In another example, a VSV sealing system or matured sealing technology is used. In an alternate example, a tube system fully sealed is used.

[0072] In the illustrated examples of FIGS. **5** and **6**, the multilayer stack **555**, **655** generates a linear displacement of the rod **560**, **660** from an electrical signal generated by an example active clearance controller **575**, **675**. An example implementation of the controller **575**, **675** that generates the electrical signal is illustrated in FIG. **14**, which is described in further detail below. In some examples, the rod **560**, **660** moves the stator nozzle **635** using the linear displacement generated by the multilayer stack **555**, **655**. In the illustrated examples of FIGS. **5** and **6**, the stator nozzle **535**, **635** and inter-stage seal **540**, **640** are connected and move together. Therefore, in the illustrated examples of FIGS. **5** and **6**, the rod **560**, **660** moves the stator nozzle **535**, **635** and inter-stage seal **540**, **640** using the linear displacement generated by the actuator **550**, **650** and multilayer **555**, **655**. In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stack **555**, **655**. For example, adding layers in the multilayer stack **555**, **655**, increase the radial movement range and muscle capability for the ACC system.

[0073] In the illustrated example of FIG. **5**, the ACC system **500** has an open clearance represented by the opening between the shroud **520** and the blade **525**. The multilayer stack **555** included in the actuator **550** controls the open clearance. In the ACC system **500**, the actuator **550** receives a first electrical signal from an example controller, and the actuator **550** provides the first electrical signal to the multilayer stack **555**. The first electrical signal causes a linear displacement of the multilayer stack **555** (e.g., each stack in the multilayer stack **555** is long and thin as seen in the example FIG. **5**). An example range of linear displacement is 200 to 300 micrometers, which translates to 10 to 15 mils for muscle capability. The linear displacement of the multilayer stack **555** moves the rod **560** upwards (e.g., away from the blade **525**). The rod **560** moves the hanger **515** and shroud **520** upwards (e.g., away from the blade **525**), which increases the open clearance.

[0074] In the illustrated examples of FIGS. **5** and **6**, the actuator **550**, **650** adjusts the clearance in two directions (e.g., shrinkage and expansion). The actuator **550**, **650** can be installed for an individual shroud (e.g., the shroud **520**, **620**), partial groups (e.g., subsets) of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an entire group or set of shrouds in

a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case **505**, **605**). The actuator **550**, **650** can also be installed for an individual stator nozzle (e.g., the stator nozzle **535**, **635**), partial groups (e.g., subsets) of stator nozzles (e.g., for groups of three stator nozzles, for groups of five stator nozzles, etc.), or for an entire group or set of stator nozzles in a turbine (e.g., the stator nozzles surrounding the 360 degree inner surface of the case **505**, **605**).

[0075] FIGS. 7A and 7B are example conversion curves **700**, **720** of new versus deteriorated blade tip conditions. The curves graph the clearance **705** of the blade as a function of pressure efficiency (also referred to as $P\eta$, P_{eta} , or pressure eta) **710**. Pressure efficiency can be defined as an assessed reference point differential from the pressure at a forward measurement divided by the reference point differential to a measured aft pressure (see equation 3). New conditions **715** are shown in the conversion curve **700** of FIG. 7A, whereas deteriorated blade conditions **725** are shown in the conversion curve **720** of FIG. 7B. As an example engine deteriorates and blade tip changes are observed, the offset from the conversion curve for new conditions occurs. This enables real-time tip loss assessment. For example, the active clearance controller **575** of FIG. 5 communicates with the forward pressure sensor **565** and aft pressure sensor **570** under new engine conditions. A reference pressure is obtained at the compressor exit (not shown). The active clearance controller calculates $P\eta$ **710** using equation (2), then uses the computed $P\eta$ **710** and a previously correlated conversion curve **700** with new conditions **715** plotted to determine the correlating amount of clearance **705**. In the example where the engine conditions are deteriorated, a different conversion curve **720** with deteriorated blade conditions **725** plotted to correlate the amount of clearance **705** and to observe the offset from the conversion curve **700** for new engines. This amount of clearance **705** is then used to control the actuators **550** with multilayer stacks **555** to subsequently drive the inter-stage seal **540** to attain the clearance **705** between the inter-stage seal **540** and the rotor **545**. In an example high power application with new engine conditions, the forward pressure is measured at 100 psia, the aft pressure is measured at 60 psia, and the reference pressure at the compressor exit is measured at 300 psia. From equation (2), $P\eta$ is calculated to be 0.83. FIG. 7A is then used to find the corresponding clearance to a $P\eta$ of 0.83. In an example high power application with deteriorated engine conditions, the forward pressure is measured at 90 psia, the aft pressure is measured at 50 psia, and the reference pressure at the compressor exit is measured at 250 psia. From equation (2), $P\eta$ is calculated to be 0.80. FIG. 7B is then used to find the corresponding clearance to a $P\eta$ of 0.80.

[0076] FIGS. 8A and 8B demonstrate example conversion curves **800**, **820** for high power and low power operation, such as ground use or in-flight use, respectively. The conversion curve **800**, **820** graphs the blade clearance **705** as a function of pressure efficiency **710** for new conditions **805**, **825** and deteriorated conditions **810**, **830**. For example, active clearance controller **675** of FIG. 6 is in communication with a forward pressure sensor **665**, an aft pressure sensor **670**, and a reference compressor exit pressure sensor (not shown). The reference pressure, forward pressure, and aft pressure are measured during ground use operation (high power operation). After computing $P\eta$ **710** by using equation (2), the active clearance controller **675** uses conversion curve **800** with high power conditions plotted for new engine conditions **805** as well as deteriorated engine conditions **810** to determine the amount of clearance **705** that correlates to the given conditions. This amount of clearance **705** is then realized by moving the actuators **650** with multilayer stacks **655**. The displacement of the stator nozzle **635** and inter-stage seal **640** are subsequently controlled by the expansion or contraction of the piezoelectric material within the multilayer stacks **655**. Given low power conditions, such as in-flight cruise conditions, an example low power conversion curve **820** is used instead to plot the $P\eta$ **710** against clearance **705** for both new engine conditions **825** and deteriorated engine conditions **830**. In an example high power application with new engine conditions, the forward pressure is measured at 100 psia, the aft pressure is measured at 60 psia, and the reference pressure at the compressor exit is measured at 300 psia. From equation (2), $P\eta$ is calculated to be 0.83. FIG. 8A is then used to find the corresponding clearance to a $P\eta$ of 0.83. In

an example high power application with deteriorated engine conditions, the forward pressure is measured at 90 psia, the aft pressure is measured at 50 psia, and the reference pressure at the compressor exit is measured at 250 psia. From equation (2), $P\eta$ is calculated to be 0.80. FIG. 8A is then used to find the corresponding clearance to a $P\eta$ of 0.80. In an example low power application with new engine conditions, the forward pressure is measured at 60 psia, the aft pressure is measured at 30 psia, and the reference pressure at the compressor exit is measured at 150 psia. From equation (2), $P\eta$ is calculated to be 0.75. FIG. 8B is then used to find the corresponding clearance to a $P\eta$ of 0.75. In an example low power application with deteriorated engine conditions, the forward pressure is measured at 50 psia, the aft pressure is measured at 20 psia, and the reference pressure at the compressor exit is measured at 120 psia. From equation (2), $P\eta$ is calculated to be 0.70. FIG. 8B is then used to find the corresponding clearance to a $P\eta$ of 0.70.

[0077] FIGS. 9A and 9B illustrate state diagrams indicating a progression of air flow and pressure of real-time inter-stage seal clearance assessment by one-point pressure measurement at a midpoint **900** or an aft point **950**. Airflow moves from a forward point upstream **920** to an aft point downstream **922**. The inter-stage seal clearance **925** is shown between the stator **910** and rotor **915**. In FIG. 9A, total pressure (P_t) **930** and static pressure (P_s) **935** are measured at the same mid-point on the stator **910**. In FIG. 9B, P_t and P_s are measured at the same aft point on the stator **910**. $P\eta$ is calculated from a reference point (a compressor exit pressure (not shown)) differential to the static pressure **935** divided by the reference point differential to the total pressure **930**. An example $P\eta$ calculation is shown below in equation 1.

$$[00001] \quad P = \frac{(P_{\text{ref}} - P_s)}{(P_{\text{ref}} - P_t)} \quad (1)$$

[0078] FIGS. 10A and 10B illustrate methods of real-time inter-stage seal clearance assessment by two-point pressure measurement at a forward-mid measurement system **1000** or a forward-aft point measurement system **1050**. Airflow moves from a forward point upstream **1020** to an aft point downstream **1022**. The inter-stage seal clearance **1025** is shown between the stator **1010** and rotor **1015**. In FIG. 10A, forward pressure (P_{fwd}) **1030** and midpoint pressure (P_{mid}) **1035** are measured on the stator **1010**. In FIG. 9B, P_{fwd} and the aft pressure are measured on the stator **910**. $P\eta$ is calculated from a reference point (a compressor exit pressure (not shown)) differential to the forward pressure **1030** divided by the reference point differential to the aft pressure **1040**. An example $P\eta$ calculation is shown below in equation 2.

$$[00002] \quad P = \frac{(P_{\text{ref}} - P_{\text{fwd}})}{(P_{\text{ref}} - P_{\text{aft}})} \quad (2)$$

[0079] FIGS. 11A and 11B illustrate methods of real-time seal clearance assessment by three-point pressure measurement system **1100**, **1150** at a forward point **1130**, midpoint **1135** and an aft point **1140**. Airflow moves from a forward point upstream **1120** to an aft point downstream **1122**. The inter-stage seal clearance **1125** is shown between the stator **1110** and rotor **1115**. $P\eta$ is calculated from the forward point **1130** differential to the midpoint **1135** divided by the forward point **1130** differential to the aft pressure **1140**. An example $P\eta$ calculation is shown below in equation 3.

$$[00003] \quad P = \frac{(P_{\text{fwd}} - P_{\text{mid}})}{(P_{\text{fwd}} - P_{\text{aft}})} \quad (3)$$

[0080] The pressure measurement methods described in FIGS. 9A to 11B can be combined with the illustrated example HPT systems of FIGS. 5 and 6, or the example LPT systems of FIGS. 12 and 13. The single point measurement method provides a simple design with one piece of instrumentation and variation of the installation location, but the total static pressure instrumentation is a complex design and life limit. For the two point measurement method, two pieces of instrumentation are used, but the flexibility for instrumentation exists. The static pressure measurement using a conventional transducer is designed to last longer than the single point measurement method. In the three point measurement method, a greater complexity of instrumentation installation is involved, but three probes yield a greater amount of data and improve the accuracy of the measurements as relative to the single and two point measurement

methods.

[0081] FIG. 12 is a schematic cross-sectional view of an example LPT implementation of an ACC system 1200 in accordance with the teachings disclosed herein. FIG. 12 is a combination of a conventional ACC tip clearance control system and an interstage seal piezo system. The example system 1200 includes ACC 1205, engine case 1210, honeycomb shroud 1220, heat shields 1215, airfoils 1225, stator nozzle 1230, pressure sensors 1250, 1255, 1260, 1265, actuator 1235 with multilayer stack 1240, and rod 1245. In this example, two pressure sensors 1250, 1255 are used for the tip control, but are used in open loop control with feedback of the actual measurement of clearance. To contrast the pressure sensors used for tip control, two additional pressure sensors 1260, 1265 are used for inter-stage seal control. The heat shield 1215 is used to isolate the engine case 1210 from heat resultant from hot gas. In this example, the pressure sensors collect pressure data and send the collected pressure data to an active clearance controller 1205. The active clearance controller 1270 includes an open loop and/or closed loop controller, as previously discussed. The controller 1270 integrates the sensor data to send a signal to actuator 1235 and example multilayer stack 1240 to expand or contract the piezoelectric material within the multilayer stack 1240 to move the rod 1245, and subsequently the stator nozzle 1230, radially inward and outward in substantially real-time. For example, the active clearance controller 1270 measures the forward pressure and aft pressure from the forward pressure sensor 1260 and aft pressure sensor 1265 respectively.

[0082] FIG. 13 is a schematic cross-sectional view of a second example LPT implementation of an ACC system 1300 in accordance with the teachings disclosed herein. The example system 1300 includes an ACC 1370, an engine case 1310, a honeycomb shroud 1320, heat shields 1315, airfoils 1325, a stator nozzle 1330, pressure sensors 1350, 1355, 1360, 1365, an actuator 1335 with a multilayer stack 1340, and a rod 1345. In this example, the pressure sensors collect pressure data and send it to the active clearance controller 1370. The active clearance controller 1370 includes an open loop and/or closed loop controller as previously discussed. The controller integrates the sensor data to send a signal to the actuator 1335 and the multilayer stack 1340 to expand or contract the piezoelectric material within the multilayer stack to move the rod 1345, and subsequently the stator nozzle 1330, radially inward and outward. In this example, there is a second actuator 1335 and multilayer stack 1340 connected to a second rod 1345 and the shroud 1320. In this example the second actuator 1335 controls the airfoil tip clearance to maximize engine performance.

[0083] FIG. 14 is a block diagram of an example controller 1400 of an example ACC system such as the ACC systems 500, 600, 1200, 1300 in FIGS. 5, 6, 12 and 13 in accordance with the examples disclosed herein. In FIG. 14, the controller 1400 can be a full-authority digital engine control (FADEC) unit, an engine control unit (ECU), an electronic engine control (EEC) unit, etc., and/or any other type of data acquisition and/or control computing device, processor platform (e.g., processor-based computing platform), etc. The controller 1400 communicates with the example pressure sensor(s) 665, 670, engine sensor(s) 1410, and actuator(s) 650. The controller 1400 includes an example clearance determination circuitry.

[0084] In the illustrated example of FIG. 14, the controller 1400 receives values for a plurality of input variables relating to flight condition (e.g., air density, throttle lever position, engine temperatures, engine pressures, direct clearance measurements, indirect clearance measurements, etc.). The controller 1400 receives the flight condition data from the engine sensor(s) 1410. The engine sensor(s) 1410 can be mounted on the gas turbine engine 100 and/or positioned elsewhere in the aircraft (e.g., on wing, in cockpit, in main cabin, in engine compartment, in cargo, etc.). The communication between the controller 1400 and the engine sensor(s) 1410 can be one-way communication and/or two-way communication, for example. The controller 1400 computes engine operating parameters such as fuel flow, stator nozzle position, air bleed valve position, etc., using the flight condition data. The controller 1400 also gets feedback data from pressure sensors 665 and 670. For example, the controller 1400 is constantly monitoring engine conditions through

engine sensors **1410**. The engine sensors indicate the operational parameters of the engine, such as the altitude, temperature, air speed, etc. The controller **1400** computes the fuel flow, stator nozzle position and other operating parameters with the data received from the engine sensors **1410**. The controller **1400** integrates all of the data and leverages the clearance determination circuitry **1405** to actuate the actuators **650** that are connected.

[0085] In the illustrated example of FIG. **14**, the clearance determination circuitry **1405** obtains the sensor data from the example engine sensor(s) **1410**. The sensor data includes the flight condition data obtained from the gas turbine engine **100**. The clearance determination circuitry **1405** monitors engine conditions based on the sensor data from the engine sensor(s) **1410**. In some examples, the clearance determination circuitry **1405** determines if the turbine case is expanding or shrinking based on the engine conditions determined from the obtained flight condition data. In the illustrated example of FIG. **14**, the clearance determination circuitry **1405** generates electrical signals to the actuator(s) **650** of an ACC system. For example, the clearance determination circuitry **1405** obtains sensor data from the engine sensors **1410** regarding temperature in the engine casing. The clearance determination circuitry **1405** determines from this engine sensor data that the turbine case is shrinking. From the pressure sensor data also obtained from the pressure sensors **665**, **670**, the clearance determination circuitry **1405** calculates the $P\eta$ and correlates the $P\eta$ value to a clearance value. The clearance determination circuitry **1405** then sends electrical signals to the actuators **650** to attain the determined clearance and ensure a proper clearance is maintained.

[0086] FIG. **15** is a block diagram of an example implementation of the clearance determination circuitry **1405** of FIG. **14** to compute clearance determination calculations. The clearance determination circuitry **1405** of FIG. **15** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by programmable circuitry such as a Central Processor Unit (CPU) executing first instructions. Additionally or alternatively, the clearance determination circuitry of FIG. **15** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by (i) an Application Specific Integrated Circuit (ASIC) and/or (ii) a Field Programmable Gate Array (FPGA) structured and/or configured in response to execution of second instructions to perform operations corresponding to the first instructions. It should be understood that some or all of the circuitry of FIG. **15** may, thus, be instantiated at the same or different times. Some or all of the circuitry of FIG. **15** may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. **15** may be implemented by microprocessor circuitry executing instructions and/or FPGA circuitry performing operations to implement one or more virtual machines and/or containers.

[0087] The clearance determination circuitry **1405** includes example measurement circuitry **1510**, example conversion curve generation circuitry **1515**, example actuator control circuitry **1520**, example clearance calculation circuitry **1525**, and example data storage **1505**.

[0088] In some examples, the measurement circuitry **1510** is instantiated by programmable circuitry executing sensor measurement instructions and/or configured to perform operations such as those represented by the flowchart(s) of FIG. **16**.

[0089] In some examples, the conversion curve generation circuitry **1515** is instantiated by programmable circuitry executing conversion curve generation instructions and/or configured to perform operations such as those represented by the flowchart(s) of FIG. **16**.

[0090] In some examples, the clearance calculation circuitry **1525** is instantiated by programmable circuitry executing clearance calculation instructions and/or configured to perform operations such as those represented by the flowchart(s) of FIG. **16**.

[0091] In some examples, the actuator control circuitry **1520** is instantiated by programmable circuitry executing actuator control instructions and/or configured to perform operations such as those represented by the flowchart(s) of FIG. **16**.

[0092] In some examples, the clearance determination circuitry **1405** includes means for

determining a clearance. For example, the means for determining may be implemented by clearance determination circuitry **1405**. In some examples, the clearance determination circuitry **1405** may be instantiated by programmable circuitry such as the example programmable circuitry **1712** of FIG. **17**. For instance, the clearance determination circuitry **1405** may be instantiated by an example microprocessor executing machine executable instructions such as those implemented by at least blocks **1625**, **1630** of FIG. **16**. In some examples, the clearance determination circuitry **1405** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or FPGA circuitry configured and/or structured to perform operations corresponding to the machine readable instructions. Additionally or alternatively, the clearance determination circuitry **1405** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the clearance determination circuitry **1405** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine readable instructions and/or to perform some or all of the operations corresponding to the machine readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0093] Example measurement circuitry **1510** measures engine component data and pressure data from engine sensor(s) and pressure sensor(s) **665**, **670**. The measurement may be performed using one or more sensor(s) (e.g., a conventional static pressure sensor, an optical sensor, a laser-based sensor, a capacitive sensor, an Eddy current sensor, a microwave sensor, etc.). In some examples, the example measurement circuitry **1510** initiates a measurement at an aft, a mid, and/or a front location relative to a given clearance gap, as determined based on the direction of combustive gas airflow. In some examples, during initial testing to develop conversion curves that correlate pressure to clearance measurements, the measurement circuitry **1510** can determine when to initiate pressure measurement(s) based on a given power level (e.g., low power, high power), a specific altitude (e.g., at 35,000 feet, etc.), ambient temperature, and/or a specific flight cycle (e.g. take-off, cruise, landing, etc.). In an example, a mid-size engine has a core speed (PCN25%) in the range of 105 to 110 percent for a high power level, whereas PCN25% is between 75 and 85 percent for a low power level, and PCN25% is 85 to 105 percent for a mid level example.

[0094] The example conversion curve generation circuitry **1515** generates the conversion curves as exemplified in FIGS. **7A-8B**. The example conversion curves account for a multitude of factors such as altitude, power application, and/or flight cycle. The example conversion curves are used to determine the relationship between pressure efficiency ($P\eta$) and clearance. For example, the example conversion curve generation circuitry **1515** receives input from the engine and pressure sensor(s) **665**, **670**, **1410** and calculates a normalized pressure efficiency curve. Accordingly, a given clearance (mils) can be determined based on the pressure measurements. This enables the clearance determination circuitry **1405** to develop a conversion curve that can be used to identify offset from the curve due to blade tip loss to achieve a more accurate clearance gap adjustment in real-time from the pressure measurement data.

[0095] The example clearance calculation circuitry **1525** takes the data from the measurement circuitry **1510**, the generated conversion curves from the conversion curve generation circuitry **1515** and calculates the actual clearance. The example clearance calculation circuitry **1525** then compares the actual clearance to the target clearance, retrieved from data storage **1505**. The difference in clearances is the clearance adjustment to be made.

[0096] The example actuator control circuitry **1520** takes the clearance adjustment data from the clearance calculation circuitry **1525**. The example actuator control circuitry **1520** is in communication with the actuator **650** of FIG. **6** and sends a signal to adjust the actuator **650** with the multilayer stack **655** to contract or expand the associated clearance.

[0097] In the example in which a plurality of actuators **650** and a plurality of multilayer stacks **655** are used, the clearance calculation circuitry **1525** compensates for nozzle rocking and for the inter-

stage seal clearance, both of which are inter-dependent on a forward hanger **615a** and forward shroud **620a** clearance with respect to a blade **625**, and an aft hanger **615b** and aft shroud **620b** to a blade **625**.

[0098] The example data storage **1505** can be used to store any information associated with the example clearance determination circuitry **1405**. For example, the data storage **1505** can store pressure measurements obtained using one or more pressure sensor(s) **665**, **670**, **1410**, conversion curve(s) generated using the conversion curve generation circuitry **1515**, and/or clearance calculation circuitry **1525** output used by the actuator control circuitry **1520** to make clearance adjustments based on real-time data. The example data storage **1505** of the illustrated example of FIG. **15** is implemented by any memory, storage device and/or storage disc for storing data such as flash memory, magnetic media, optical media, etc. Furthermore, the data stored in the example data storage **1505** can be in any data format such as binary data, comma delimited data, tab delimited data, structured query language (SQL) structures, image data, etc.

[0099] While an example implementation of the clearance determination circuitry **1405** of FIG. **14** is illustrated in FIG. **15**, one or more of the elements, processes, and/or devices illustrated in FIG. **15** may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the measurement circuitry **1510**, conversion curve generation circuitry **1515**, clearance calculation circuitry **1525**, actuator control circuitry **1520**, and/or, more generally, the example clearance determination circuitry **1405** of FIG. **15**, may be implemented by hardware alone or by hardware in combination with software and/or firmware. Thus, for example, any of the measurement circuitry **1510**, conversion curve generation circuitry **1515**, clearance calculation circuitry **1525**, actuator control circuitry **1520**, and/or, more generally, the example clearance determination circuitry **1405**, could be implemented by programmable circuitry in combination with machine readable instructions (e.g., firmware or software), processor circuitry, analog circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), ASIC(s), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as FPGAs. Further still, the example clearance determination circuitry **1405** of FIG. **15** may include one or more elements, processes, and/or devices in addition to, or instead of, those illustrated in FIG. **15**, and/or may include more than one of any or all of the illustrated elements, processes and devices.

[0100] A flowchart representative of example machine readable instructions, which may be executed by programmable circuitry to implement and/or instantiate the clearance determination circuitry **1405** of FIGS. **14** and/or **15** and/or representative of example operations which may be performed by programmable circuitry to implement and/or instantiate the clearance determination circuitry **1405** of FIG. **15**, is shown in FIG. **16**. The machine readable instructions may be one or more executable programs or portion(s) of one or more executable programs for execution by programmable circuitry such as the programmable circuitry **1712** shown in the example processor platform **1700** discussed below in connection with FIG. **17** and/or may be one or more function(s) or portion(s) of functions to be performed by the example programmable circuitry (e.g., an FPGA). In some examples, the machine readable instructions cause an operation, a task, etc., to be carried out and/or performed in an automated manner in the real world. As used herein, “automated” means without human involvement.

[0101] The program may be embodied in instructions (e.g., software and/or firmware) stored on one or more non-transitory computer readable and/or machine readable storage medium such as cache memory, a magnetic-storage device or disk (e.g., a floppy disk, a Hard Disk Drive (HDD), etc.), an optical-storage device or disk (e.g., a Blu-ray disk, a Compact Disk (CD), a Digital Versatile Disk (DVD), etc.), a Redundant Array of Independent Disks (RAID), a register, ROM, a solid-state drive (SSD), SSD memory, non-volatile memory (e.g., electrically erasable programmable read-only memory (EEPROM), flash memory, etc.), volatile memory (e.g., Random

Access Memory (RAM) of any type, etc.), and/or any other storage device or storage disk. The instructions of the non-transitory computer readable and/or machine readable medium may program and/or be executed by programmable circuitry located in one or more hardware devices, but the entire program and/or parts thereof could alternatively be executed and/or instantiated by one or more hardware devices other than the programmable circuitry and/or embodied in dedicated hardware. The machine readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a human and/or machine user) or an intermediate client hardware device gateway (e.g., a radio access network (RAN)) that may facilitate communication between a server and an endpoint client hardware device. Similarly, the non-transitory computer readable storage medium may include one or more mediums. Further, although the example program is described with reference to the flowchart(s) illustrated in FIG. 16, many other methods of implementing the example clearance determination circuitry 1405 may alternatively be used. For example, the order of execution of the blocks of the flowchart(s) may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks of the flow chart may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The programmable circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core CPU), a multi-core processor (e.g., a multi-core CPU, an XPU, etc.)). For example, the programmable circuitry may be a CPU and/or an FPGA located in the same package (e.g., the same integrated circuit (IC) package or in two or more separate housings), one or more processors in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, etc., and/or any combination(s) thereof.

[0102] The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine readable instructions as described herein may be stored as data (e.g., computer-readable data, machine-readable data, one or more bits (e.g., one or more computer-readable bits, one or more machine-readable bits, etc.), a bitstream (e.g., a computer-readable bitstream, a machine-readable bitstream, etc.), etc.) or a data structure (e.g., as portion(s) of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions may be fragmented and stored on one or more storage devices, disks and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of computer-executable and/or machine executable instructions that implement one or more functions and/or operations that may together form a program such as that described herein.

[0103] In another example, the machine readable instructions may be stored in a state in which they may be read by programmable circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine-readable instructions on a particular computing device or other

device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine readable, computer readable and/or machine readable media, as used herein, may include instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s).

[0104] The machine readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine readable instructions may be represented using any of the following languages: C, C++, Java, C #, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

[0105] As mentioned above, the example operations of FIG. **16** may be implemented using executable instructions (e.g., computer readable and/or machine readable instructions) stored on one or more non-transitory computer readable and/or machine readable media. As used herein, the terms non-transitory computer readable medium, non-transitory computer readable storage medium, non-transitory machine readable medium, and/or non-transitory machine readable storage medium are expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media. Examples of such non-transitory computer readable medium, non-transitory computer readable storage medium, non-transitory machine readable medium, and/or non-transitory machine readable storage medium include optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory (ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms “non-transitory computer readable storage device” and “non-transitory machine readable storage device” are defined to include any physical (mechanical, magnetic and/or electrical) hardware to retain information for a time period, but to exclude propagating signals and to exclude transmission media. Examples of non-transitory computer readable storage devices and/or non-transitory machine readable storage devices include random access memory of any type, read only memory of any type, solid state memory, flash memory, optical discs, magnetic disks, disk drives, and/or redundant array of independent disks (RAID) systems. As used herein, the term “device” refers to physical structure such as mechanical and/or electrical equipment, hardware, and/or circuitry that may or may not be configured by computer readable instructions, machine readable instructions, etc., and/or manufactured to execute computer-readable instructions, machine-readable instructions, etc.

[0106] FIG. **16** is a flowchart representative of machine readable instructions that can be executed to implement the example active clearance controller **1400** of FIG. **15** in conjunction with the example ACC system of FIGS. **5**, **6**, **12**, and **13**. The program **1600** of FIG. **16** begins execution at block **1605**, at which the example measurement circuitry **1510** obtains sensor data from the example engine sensor(s) **1410**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1410** from an engine (e.g., the gas turbine engine **100** of FIG. **1**). In some examples, flight condition data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.). The sensor data varies depending on the operation.

[0107] At block **1610**, the example measurement circuitry **1510** obtains sensor data from the example pressure sensor(s) **665**, **670**. In some examples, the sensor data includes a singular point of pressure measurement as shown in FIGS. **9A** and **9B**, a two-point measurement system as shown in FIGS. **10A** and **10B**, or three-point measurement system as shown in FIGS. **11A** and **11B**. In some examples, a reference pressure is taken at a compressor outlet. For example, reference pressure is measured using a pressure sensor at the compressor outlet.

[0108] At block **1615**, the conversion curve generation circuitry **1515** receives sensor-based input data and determines a correlation between the normalized pressure efficiency ($P\eta$) and the blade clearance. The correlation between the normalized pressure efficiency and the blade clearance is used to generate a conversion curve by the conversion curve generation circuitry **1515**. The conversion curve generation is performed at block **1620**. For example, testing may be performed in order to collect data to create initial correlations. In some examples, the conversion curve generation circuitry **1515** can generate conversion curves for a range of test flights, for a new engine, or for an engine at various flight cycles. The collection of data for conversion curve generation enables the conversion curves to be validated and the observation and/or testing of engines with gradual blade loss to investigate the effects of blade length changes on pressure efficiency measurements. In some examples, the testing can be performed at varying power levels (low power, high power, etc.), as well as a range of altitudes. An example range of altitudes is 5,000 ft for low altitude up to 35,000 ft for high altitude. Thorough testing and conversion curve development permits the usage of the clearance determination circuitry **1405** during actual in-flight monitoring of clearances and contributes to a more accurate adjustment of the clearances by the ACC **1400**.

[0109] At block **1625**, the example conversion curve generation circuitry **1515** determines a correlation between the normalized pressure efficiency and the blade clearance. This correlation is used by the conversion curve generation circuitry **1515** to generate a conversion curve. The conversion curve is generated for a given power level and/or altitude, based on pressure data and obtained data for a range of test flights performed to observe blade tip loss.

[0110] Once the conversion curves have been generated for various power levels and/or altitudes, the measurement circuitry **1510** measures the real-time pressure measurements and feeds that data to the clearance calculation circuitry **1525**. At block **1625**, the clearance calculation circuitry **1525** uses the conversion curve generated to determine the real-time clearance. At block **1630**, the blade tip loss is determined based on the off-set of the curve and the interpolation between the curves. For example, a conversion curve is generated from simulations for a new engine. During real-time, in-flight data collection, pressure measurements are taken and used to calculate pressure efficiency. Expected pressure measurements are generated from initial test conditions. Deviation from the expected pressure measurements indicates blade tip loss, or off-set from the curve.

[0111] Once the blade tip loss data is calculated based on the off-set of the conversion curve at block **1630**, the blade tip loss data is input to the active clearance controller **1400** for storage and later usage at block **1635**. At block **1640**, the actuator control circuitry **1520** uses the real-time clearance and associated blade tip loss to send an electrical signal to the actuator **550**, **650**, **1235**, **1335** and multilayer stack **555**, **655**, **1240**, **1340** to actuate the rods of FIGS. **5**, **6**, **12**, and **13** and adjust clearance of the stator nozzle **535**, **635**, **1230**, **1330** and inter-stage seal **540**, **640**.

[0112] FIG. **17** is a block diagram of an example programmable circuitry platform **1700** structured to execute and/or instantiate the example machine-readable instructions and/or the example operations of FIG. **16** to implement the clearance determination circuitry **1405** of FIG. **14**. The programmable circuitry platform **1700** can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), or any other type of computing and/or electronic device.

[0113] The programmable circuitry platform **1700** of the illustrated example includes programmable circuitry **1712**. The programmable circuitry **1712** of the illustrated example is hardware. For example, the programmable circuitry **1712** can be implemented by one or more integrated circuits, logic circuits, FPGAs, microprocessors, CPUs, GPUs, DSPs, and/or microcontrollers from any desired family or manufacturer. The programmable circuitry **1712** may be implemented by one or more semiconductor based (e.g., silicon based) devices. In this example, the programmable circuitry **1712** implements the clearance determination circuitry **1405**.

[0114] The programmable circuitry **1712** of the illustrated example includes a local memory **1713**

(e.g., a cache, registers, etc.). The programmable circuitry **1712** of the illustrated example is in communication with main memory **1716**, which includes a volatile memory **1714** and a non-volatile memory **1716**, by a bus **1718**. The volatile memory **1714** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory **1716** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1714**, **1716** of the illustrated example is controlled by a memory controller **1717**. In some examples, the memory controller **1717** may be implemented by one or more integrated circuits, logic circuits, microcontrollers from any desired family or manufacturer, or any other type of circuitry to manage the flow of data going to and from the main memory **1714**, **1716**.

[0115] The programmable circuitry platform **1700** of the illustrated example also includes interface circuitry **1720**. The interface circuitry **1720** may be implemented by hardware in accordance with any type of interface standard, such as an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.

[0116] In the illustrated example, one or more input devices **1722** are connected to the interface circuitry **1720**. The input device(s) **1722** permit(s) a user (e.g., a human user, a machine user, etc.) to enter data and/or commands into the programmable circuitry **1712**. The input device(s) **1722** can be implemented by, for example, a pressure sensor, a temperature sensor, etc.

[0117] One or more output devices **1724** are also connected to the interface circuitry **1720** of the illustrated example. The output device(s) **1724** can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube (CRT) display, an in-place switching (IPS) display, a touchscreen, etc.), etc. The interface circuitry **1720** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip, and/or graphics processor circuitry such as a GPU.

[0118] The interface circuitry **1720** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) by a network **1726**. The communication can be by, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a beyond-line-of-sight wireless system, a line-of-sight wireless system, a cellular telephone system, an optical connection, etc.

[0119] The programmable circuitry platform **1700** of the illustrated example also includes one or more mass storage discs or devices **1728** to store firmware, software, and/or data. Examples of such mass storage discs or devices **1728** include magnetic storage devices (e.g., floppy disk, drives, HDDs, etc.), optical storage devices (e.g., Blu-ray disks, CDs, DVDs, etc.), RAID systems, and/or solid-state storage discs or devices such as flash memory devices and/or SSDs.

[0120] The machine readable instructions **1732**, which may be implemented by the machine readable instructions of FIGS. **16**, may be stored in the mass storage device **1728**, in the volatile memory **1714**, in the non-volatile memory **1716**, and/or on at least one non-transitory computer readable storage medium such as a CD or DVD which may be removable.

[0121] As shown in FIG. **17**, the measurement circuitry **1510**, conversion curve generation circuitry **1515**, actuator control circuitry **1520**, and clearance calculation circuitry may be instantiated in the programmable circuitry **1712**. The data storage may be instituted by a local memory **1713**, volatile memory **1714**, non-volatile memory **1716**, or mass storage device **1728**.

[0122] FIG. **18** shows that in addition to using piezoelectric materials or SMA materials with a multilayer stack, a mechanical arm could be used in conjunction with an actuator **1850**. FIG. **18** illustrates an example ACC system **1800**, which can be used as a mechanical arm actuator. The

ACC system **1800** includes an active clearance controller **1875**, an engine case **1805**, multiple adjustment sleeves **1807**, guide hooks **1810A**, **1810B**, a forward hanger **1815a**, a forward shroud **1820a**, an aft hanger **1815b**, an aft shroud **1820b**, a stator nozzle **1835**, a mid-stage seal **1817**, an inter-stage seal **1840**, a rotor **1845**, blades **1825**, a forward pressure sensor **1865**, an aft pressure sensor **1870**, an annular sync ring **1848**, a sync arm **1846**, and an externally threaded rod **1860** having an outboard end **1844**.

[0123] For example, an active clearance controller **1875** and actuator **1850** may be connected to an annular sync ring **1848** having a sync arm **1846**. The example sync arm **1846** is connected to a first end of an externally threaded rod **1860** with an outboard end **1844**. The externally threaded rod **1860** is threaded through an adjustment sleeve **1807** with internal and external threading located inside a threaded internal bore in an example engine case **1805**. A second end of the externally threaded rod **1860** is connected to an example stator nozzle **1835**. Angular rotation of the sync ring **1848** causes a pivoting movement of the sync arm **1846**. The components are coupled to enable rotation of the externally threaded rod with the pivoting movement of the sync arm **1846**, causing the externally threaded rod to move radially inward or outward, and adjust the coupled stator nozzle **1835**. In turn, the movement of the rod **1860** and stator nozzle **1835** affects the clearance of an inter-stage seal **1840** to an example rotor **1845**. An example active clearance controller **1875** is connected to a forward pressure sensor **1865** and an aft pressure sensor **1870**. Using the methods disclosed herein, the active clearance controller **1875** leverages conversion curves to determine the clearance between the inter-stage seal **1840** and the rotor **1845**. The active clearance controller **1875** accounts for expansion of the engine case **1805** due to heat, the displacement of the forward hanger **1815** and shroud **1820**, the aft hanger **1815** and shroud **1820**, as well as the determined clearance of the stator nozzle **1835** and inter-stage seal **1840** from the rotor **1845**. An electrical signal is sent to the actuator **1850** which actuates the annular sync ring **1848**. The annular sync ring **1848** rotates, causing pivoting of the sync arm **1846**, which is connected to a first end of the externally threaded rod **1860**. The externally threaded rod **1860** rotates, so the internally and externally threaded sleeve **1807** within the engine case **1805**, allows the rod to adjust radially inward or outward to adjust the displacement of the stator nozzle **1835** and the inter-stage seal **1840**. By adjusting the displacement of the inter-stage seal **1840**, the active clearance controller **1875** has finite, real-time control over the clearance. In an example real-world muscle range, a range greater than 15 mils for a mid-size engine is used, whereas a range greater than 30 mils is used for a large size engine.

[0124] From the foregoing, it will be appreciated that example methods, apparatus and articles of manufacture have been disclosed that improve clearance control of inter stage and mid stage seals in a gas turbine engine. The disclosed examples propose improved ACC designs using a combination of piezoelectric actuators, pressure measurements, and an active clearance controller to achieve tighter clearance at any operating conditions with rapid mechanical ACC modulation. The disclosed examples use piezoelectric material to generate high mechanical power and provide rapid response clearance control in two directions (inward and outward) with no time delay. The disclosed examples use multilayer stacks of the piezoelectric material to manage the range of displacement, which affects the range of the ACC system muscle capability. The disclosed examples propose real-time clearance assessment methods and rapid response to improve efficiency and operability of engine conditions, to control inter-stage seal clearance, and to prevent nozzle rocking.

[0125] Example methods, apparatus, systems, and articles of manufacture to provide rapid active clearance control of inter-stage and mid-stage seals are disclosed herein. Further examples and combinations thereof include the following:

[0126] An apparatus to control clearance for a turbine engine, the apparatus comprising a case surrounding at least part of the turbine engine, the at least part of the turbine engine including a nozzle, the nozzle to contain airflow, the nozzle including a reference pressure sensor at a first

location on a tip of the nozzle and a static pressure sensor at a second location on the tip of the nozzle, a first actuator to control a clearance between a rotor and the nozzle, the first actuator including a multilayer stack of material, wherein the first actuator is positioned at a third location outside of the case, a first rod coupled to the first actuator and the nozzle through an opening in the case at the third location, the rod to move the nozzle based on contraction or expansion of the multilayer stack of material, and a controller to calculate and set the clearance between the rotor and the nozzle by supplying an electrical current to the multilayer stack in the first actuator.

[0127] The apparatus of any preceding clause, wherein the multilayer stack of material is a first multilayer stack of material, wherein the opening is a first opening, and further including a second actuator to control the clearance between a blade and the nozzle, the second actuator including a second multilayer stack of material, wherein the second actuator is positioned at a fourth location outside of the case, and a second rod coupled to the second actuator and the nozzle through a second opening in the case at the fourth location, the second rod to move the nozzle based on the second multilayer stack of material.

[0128] The apparatus of any preceding clause, wherein the controller uses a closed loop system, the closed loop system to set a target clearance, the controller to manipulate the actuator to achieve the target clearance based on a clearance calculation, the clearance calculation including at least an engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure.

[0129] The apparatus of any preceding clause, wherein the reference and static pressure measurements are used to develop a normalized pressure measurement, the normalized pressure measure used to generate a conversion curve correlating the normalized pressure measurement with a clearance measurement, the conversion curve used to compare to real-time pressure measurements to adjust a blade tip clearance.

[0130] The apparatus of any preceding clause, wherein the controller uses an open loop system, the open loop system to set a target clearance based on the conversion curve and the normalized pressure measurement, the open loop system to manipulate the actuator to achieve the target clearance.

[0131] The apparatus of any preceding clause, wherein the multilayer stack of material includes at least one of piezoelectric material or shape memory alloy.

[0132] The apparatus of any preceding clause, wherein the reference pressure sensor is placed on at least one of a compressor exit location, an aft location on the nozzle, a middle location on the nozzle, or a forward location on the nozzle.

[0133] The apparatus of any preceding clause, wherein the static pressure sensor is placed on at least one of an aft location, a middle location, or a forward location.

[0134] The apparatus of any preceding clause, wherein the conversion curve is generated during testing at a plurality of altitudes.

[0135] The apparatus of any preceding clause, wherein the conversion curve is generated during testing at a plurality of power levels, the plurality of power levels including at least one of a low power or a high power.

[0136] An apparatus to control clearance in a turbine engine, the apparatus comprising a controller to determine a first target clearance between a rotor and a nozzle as well as a second target clearance between a blade and a shroud, the controller in communication with at least one pressure sensor, the control system to supply electrical power to a plurality of actuators, a first actuator to actuate a first rod to achieve the first target clearance based on the controller determination, the first actuator located outside of an engine case and coupled to a first end of the first rod, the first rod positioned through a first seal in the engine case, a second end of the first rod coupled to the nozzle, and a second actuator to actuate a second rod to achieve the second target clearance based on the controller determination, the second actuator located outside of the engine case and coupled to a first end of the second rod, the second rod positioned through a second seal in the engine case,

a second end of the second rod coupled to a hanger from which the shroud hangs.

[0137] The apparatus of any preceding clause, wherein the controller uses a closed loop system, the closed loop system to determine the first and second target clearances based on a calculation including an engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure.

[0138] The apparatus of any preceding clause, wherein the controller uses an open loop system, the open loop system to set the first and second target clearances based on a conversion curve and a normalized pressure measurement.

[0139] The apparatus of any preceding clause, wherein the normalized pressure measurement is developed from reference and static pressure measurements, the normalized pressure measurement used to generate the conversion curve correlating the normalized pressure measurement with a clearance measurement, the conversion curve used to compare to real-time pressure measurements to achieve the first and second target clearances.

[0140] The apparatus of any preceding clause, wherein the conversion curve is generated during testing at a plurality of altitudes.

[0141] The apparatus of any preceding clause, wherein the conversion curve is generated during testing at a plurality of power levels, the plurality of power levels including at least one of a low power or a high power.

[0142] The apparatus of any preceding clause, wherein the conversion curve is generated during testing for a multitude of flight cycles.

[0143] The apparatus of any preceding clause, wherein the actuator includes a multilayer stack of material including at least one of piezoelectric material or shape memory alloy.

[0144] The apparatus of any preceding clause, wherein the at least one pressure sensor measures pressure at one of a compressor exit location, a forward location on the nozzle, an aft location on the nozzle, or a mid-point location on the nozzle.

[0145] An apparatus to control clearance for a turbine engine, the apparatus comprising a case surrounding at least part of the turbine engine, the at least part of the turbine engine including a nozzle, the nozzle to contain airflow, the nozzle including a reference pressure sensor at a first location on a tip of the nozzle and a static pressure sensor at a second location on the tip of the nozzle, a mechanical arm to control a clearance between a rotor and the nozzle, wherein the mechanical arm is positioned at a third location outside of the case, a rod coupled to an actuator and the nozzle through an opening in the case at the third location, the rod to move the nozzle based on contraction or expansion of a multilayer stack of material, and a controller to calculate and set the clearance between the rotor and the nozzle by supplying an electrical current to the multilayer stack in the actuator.

[0146] The apparatus of any preceding clause, wherein the controller uses a closed loop system, the closed loop system to determine the first and second target clearances based on a calculation including an engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure.

[0147] The apparatus of any preceding clause, wherein the controller uses an open loop system, the open loop system to set the first and second target clearances based on a conversion curve and a normalized pressure measurement.

[0148] The apparatus of any preceding clause, wherein the normalized pressure measurement is developed from reference and static pressure measurements, the normalized pressure measurement used to generate the conversion curve correlating the normalized pressure measurement with a clearance measurement, the conversion curve used to compare to real-time pressure measurements to achieve the first and second target clearances.

[0149] The apparatus of any preceding clause, wherein the conversion curve is generated during testing at a plurality of altitudes.

[0150] The apparatus of any preceding clause, wherein the conversion curve is generated during

testing at a plurality of power levels, the plurality of power levels including at least one of a low power or a high power.

[0151] The apparatus of any preceding clause, wherein the conversion curve is generated during testing for a multitude of flight cycles.

[0152] The apparatus of any preceding clause, wherein the actuator includes a multilayer stack of material including at least one of piezoelectric material or shape memory alloy.

[0153] The apparatus of any preceding clause, wherein the at least one pressure sensor measures pressure at one of a compressor exit location, a forward location on the nozzle, an aft location on the nozzle, or a mid-point location on the nozzle.

[0154] A system to control clearance in a turbine engine, the system comprising at least one actuator to house a multilayer stack of material that moves a rod in a radial direction, the rod coupled to a nozzle, at least one sensor placed inside of an engine case, and a controller to use the at least one sensor to monitor a parameter, convert the monitored parameter into an actual clearance between a tip of the nozzle and a rotor, determine the difference between the actual clearance and a predetermined target clearance, supply electrical power to the multilayer stack to actuate the rod to achieve the target clearance.

[0155] The system of any preceding clause, wherein the sensor is a pressure sensor placed on the tip of the nozzle within the engine case.

[0156] The system of any preceding clause, wherein the sensor is a pressure sensor placed on a shroud within the engine case.

[0157] The system of any preceding clause, wherein the predetermined target clearance is obtained from test data, the test data collected for a variety of flight cycles, altitudes, and engine power levels.

[0158] The system of any preceding clause, wherein the actuator is located outside of the engine case.

[0159] The system of any preceding clause, wherein the multilayer stack includes at least one of a piezoelectric material or a smart metal alloy.

[0160] The system of any preceding clause, wherein the normalized pressure measurement includes a difference between a reference pressure and a static pressure, divided by the difference between the reference pressure and a total pressure.

[0161] The system of any preceding clause, wherein the normalized pressure measurement includes a difference between a reference pressure and a forward pressure, divided by the difference between the reference pressure and an aft pressure.

[0162] The system of any preceding clause, wherein the normalized pressure measurement includes a difference between a static pressure and a reference pressure, divided by the difference between a total pressure and a reference pressure.

[0163] The system of any preceding clause, wherein the normalized pressure measurement includes a difference between a forward point pressure and a midpoint pressure, divided by the difference between the forward pressure and an aft pressure.

[0164] The system of any preceding clause, wherein the controller is an active clearance controller comprising clearance determination circuitry, the clearance determination circuitry connected to an engine sensor, a pressure sensor, and an actuator.

[0165] The system of any preceding clause, wherein the clearance determination circuitry further includes measurement circuitry, conversion curve generation circuitry, clearance calculation circuitry, actuator control circuitry, and data storage.

[0166] The apparatus of any preceding clause, wherein the normalized pressure measurement includes a difference between a reference pressure and a static pressure, divided by the difference between the reference pressure and a total pressure.

[0167] The apparatus of any preceding clause, wherein the normalized pressure measurement includes a difference between a reference pressure and a forward pressure, divided by the

difference between the reference pressure and an aft pressure.

[0168] The apparatus of any preceding clause, wherein the normalized pressure measurement includes a difference between a static pressure and a reference pressure, divided by the difference between a total pressure and a reference pressure.

[0169] The apparatus of any preceding clause, wherein the normalized pressure measurement includes a difference between a forward point pressure and a midpoint pressure, divided by the difference between the forward pressure and an aft pressure.

[0170] The apparatus of any preceding clause, wherein the controller is an active clearance controller comprising clearance determination circuitry, the clearance determination circuitry connected to an engine sensor, a pressure sensor, and an actuator.

[0171] The apparatus of any preceding clause, wherein the clearance determination circuitry further includes measurement circuitry, conversion curve generation circuitry, clearance calculation circuitry, actuator control circuitry, and data storage.

[0172] Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. For example, the disclosed example methods, apparatus and articles of manufacture are implemented in conjunction with a gas turbine engine, however, the disclosed examples can be implemented in conjunction with a compressor. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

[0173] The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Claims

1. An apparatus to control clearance for a turbine engine, the apparatus comprising: a case surrounding at least part of the turbine engine; a nozzle to contain airflow, the nozzle including a first pressure sensor at a first location on a tip of the nozzle and a second pressure sensor at a second location on the tip of the nozzle; an actuator to control a clearance between a rotor and the nozzle, wherein the actuator is positioned at a third location outside of the case; a rod coupled to the actuator and the nozzle through an opening in the case at the third location, the rod to move the nozzle based on contraction or expansion of a multilayer stack of material; and a controller to calculate and set the clearance between the rotor and the nozzle by supplying an electrical current to the multilayer stack in the actuator.
2. The apparatus of claim 1, wherein a third pressure sensor and a fourth pressure sensor are located between the case and a tip of the rotor.
3. The apparatus of claim 2, wherein the third pressure sensor and the fourth pressure sensor provide feedback via an open loop feedback system to the controller to set a tip clearance between the case and the tip of the rotor.
4. The apparatus of claim 3, wherein the tip clearance is measured between a honeycomb shroud and the tip of the rotor, wherein the honeycomb shroud is located between the case and the tip of the rotor.
5. The apparatus of claim 1, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the nozzle via a closed loop feedback system.
6. The apparatus of claim 1, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the nozzle via an open loop feedback system.
7. The apparatus of claim 1, wherein the controller uses a closed loop feedback system, the closed loop feedback system to set a target clearance, the controller to manipulate the actuator to achieve the target clearance based on a clearance calculation, the clearance calculation including at least an

engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure.

8. A turbine engine, the turbine engine comprising: a case surrounding at least part of the turbine engine; a nozzle to contain airflow, the nozzle including a first pressure sensor at a first location on a tip of the nozzle and a second pressure sensor at a second location on the tip of the nozzle; an actuator to control a clearance between a rotor and the nozzle, wherein the actuator is positioned at a third location outside of the case; a rod coupled to the actuator and the nozzle through an opening in the case at the third location, the rod to move the nozzle based on contraction or expansion of a multilayer stack of material; and a controller to calculate and set the clearance between the rotor and the nozzle by supplying an electrical current to the multilayer stack in the actuator.

9. The turbine engine of claim 8, wherein a third pressure sensor and a fourth pressure sensor are located between the case and a tip of the rotor.

10. The turbine engine of claim 9, wherein the third pressure sensor and the fourth pressure sensor provide feedback via an open loop feedback system to the controller to set a tip clearance between the case and the tip of the rotor.

11. The turbine engine of claim 10, wherein the tip clearance is measured between a honeycomb shroud and the tip of the rotor, wherein the honeycomb shroud is located between the case and the tip of the rotor.

12. The turbine engine of claim 8, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the nozzle via a closed loop feedback system.

13. The turbine engine of claim 8, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the nozzle via an open loop feedback system.

14. The turbine engine of claim 8, wherein the controller uses a closed loop feedback system, the closed loop feedback system to set a target clearance, the controller to manipulate the actuator to achieve the target clearance based on a clearance calculation, the clearance calculation including at least an engine speed, a turbine pressure, a turbine temperature, a compressor temperature, and a compressor pressure.

15. An apparatus to control clearance for a turbine engine, the apparatus comprising: a means for surrounding at least part of the turbine engine; a means for containing airflow including a first pressure sensor at a first location on a tip of the means for containing and a second pressure sensor at a second location on the tip of the means for containing; a means for controlling a clearance between a rotor and the means for containing, wherein the means for controlling is positioned at a third location outside the means for surrounding; a means for moving the means for containing based on contraction or expansion of a multilayer stack of material, the means for moving coupled to the means for controlling and the means for containing through an opening in the means for surrounding at the third location; and a means for calculating and setting the clearance between the rotor and the means for containing by supplying an electrical current to the multilayer stack in the means for controlling.

16. The apparatus of claim 15, wherein a third pressure sensor and a fourth pressure sensor are located between the means for surrounding and a tip of the rotor.

17. The apparatus of claim 16, wherein the third pressure sensor and the fourth pressure sensor provide feedback via an open loop feedback system to the means for controlling to set a tip clearance between the means for surrounding and the tip of the rotor.

18. The apparatus of claim 17, wherein the tip clearance is measured between a honeycomb shroud and the tip of the rotor, wherein the honeycomb shroud is located between the means for surrounding and the tip of the rotor.

19. The apparatus of claim 15, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the means for containing via a

closed loop feedback system.

20. The apparatus of claim 15, wherein the first pressure sensor and the second pressure sensor provide a measurement of the clearance between the rotor and the means for containing via an open loop feedback system.
